

C04

Prgm1

Program:

```
class Rectangle:
    def __init__(self,l,b, cost=0):
        self.length =l
        self.breadth =b

    def perimeter(self):
        return 2 *(self.length +self.breadth)
    def area(self):
        return self.length *self.breadth
    def compare(self,r2):
        if r2.area() > r.area():
            print("Area of rectangle 2 is greater")
        elif r2.area() < r.area():
            print("Area of rectangle 1 is greater")
        else:
            print("Both area are Equal")

        if r2.perimeter() > r.perimeter():
            print("perimeter of rectangle 2 is greater")
        elif r2.perimeter() < r.perimeter():
            print("perimeter of rectangle 1 is greater")
        else:
            print("Both perimeter are Equal")
r=Rectangle(10,20)
r2=Rectangle(10,10)
print("Area of 1st rectangle:",r.area())
print("Perimeter of 1st rectangle:",r.perimeter())
print()
print("Area of 2nd rectangle:",r2.area())
print("Perimeter of 2nd rectangle:",r2.perimeter())
print()
r.compare(r2)
```

Output:

Area of 1st rectangle: 200
Perimeter of 1st rectangle: 60

Area of 2nd rectangle: 100
Perimeter of 2nd rectangle: 40

Area of rectangle 1 is greater
perimeter of rectangle 1 is greater

Prgm3

Program:

```
class rectangle:
    def __init__(self):
        self.__length=int(input("length:"))
        self.__breadth=int(input("breadth:"))
    def __lt__(self,ob2):
        area1=self.__length * self.__breadth
        area2=ob2.__length *ob2.__breadth
        print("the area1:",area1)
        print("the area2:",area2)
        return (area1<area2)
print("Rectangle-1")
og1=rectangle()
print("Rectangle-2")
og2=rectangle()

print(og1<og2)

if og1<og2:
    print("rectangle 1 is smaller")
else:
    print("rectangle 2 is smaller")
```

Output:

Rectangle-1
length:5
breadth:4
Rectangle-2
length:9
breadth:2
the area1: 20
the area2: 18
area1<area2: False
the area1: 20
the area2: 18
rectangle 2 is smaller

Prgm2

Program:

class BankAccount:

```
    def
__init__(self,AccNo,AccName,AccType,AccBalance
=0):
    self.number=AccNo
    self.Name=AccName
    self.Atype=AccType
    self.Balance=AccBalance
def withdraw(self,Amount):
    if( Amount>=self.Balance ):
        print("Low balance")
    else:
        self.Balance= self.Balance-Amount
def deposit(self,Amount):
    self.Balance= self.Balance+Amount
def display(self):
    print("Account Number: ",self.number)
    print("Account Name: ",self.Name)
    print("Account type: ",self.Atype)
    print("Account Balance: ",self.Balance)
```

accno=int(input("Enter your acc No:"))

name=input("Enter your Name:")

obj =BankAccount(accno, name, "Savings")

print("\n1.Account

info\n2.Deposit\n3.Withdraw\n4.Exit")

while(1):

```
    opt=int(input("Select your option:"))
```

```
    if opt == 1:
```

```
        obj.display()
```

```
    elif opt == 2:
```

```
        dep=int(input("Enter the value to deposite: "))
```

```
        obj.deposit(dep)
```

```
        print("Account Balance: ",obj.Balance)
```

```
    elif opt == 3:
```

```
        wid=int(input("Enter the value to withdraw:
```

```
"))
```

```
        obj.withdraw(wid)
```

```
        print("Account Balance: ",obj.Balance)
```

```
    elif opt == 4:
```

```
        print("Exit")
```

```
        break
```

```
    else:
```

```
        print("Invalid Option")
```

Output:

Enter your acc No:987654321

Enter your Name:Jane

1.Account info

2.Deposit

3.Withdraw

4.Exit

Select your option:1

Account Number: 987654321

Account Name: Jane

Account type: Savings

Account Balance: 0

Select your option:2

Enter the value to deposite: 3000

Account Balance: 3000

Select your option:3

Enter the value to withdraw: 510

Account Balance: 2490

Select your option:1

Account Number: 987654321

Account Name: Jane

Account type: Savings

Account Balance: 2490

Select your option:4

Exit

Prgm5

Program:

```
class publisher:
    def __init__(self,n):
        self.name=n
    def display(self):
        print("Name of the book is",self.name)
class book(publisher):
    def __init__(self,n,t,a):
        super().__init__(n)
        self.title=t
        self.author=a
    def display(self):
        print("Title of the book is",self.title)
        print("Author of the book is",self.author)
class python(book):
    def __init__(self,n,t,a,p,pgs):
        super().__init__(n,t,a)
        self.price=p
        self.pages=pgs
    def display(self):
        print("Name of the book is:",self.name)
        print("Title of the book is:",self.title)
        print("Author of the book is:",self.author)
        print("Price of the book is :",self.price)
        print("No of pages:",self.pages)
p1=python("python","Introduction to
python","Jeeva Jose",450,300)
p1.display()
```

Output:

Name of the book is: python
Title of the book is: Introduction to python
Author of the book is: Jeeva Jose
Price of the book is : 450
No of pages: 300

Prgm4

Program:

```
class Time:
    def __init__(self):
        self.__h=int(input("Enter the hour:"))
        self.__m=int(input("Enter the minute:"))
        self.__s=int(input("Enter the second:"))
    def __add__(self,time2):
        hour =self.__h + time2.__h

        minute =self.__m + time2.__m

        second =self.__s + time2.__s
        if hour >=24:
            hour=hour%24
        if minute >=60:
            hour =hour+int(minute/60)
            minute=minute%60
        if second >=60:
            minute =minute + int(second/60)
            second =second%60
        #return second,minute,hour
        print("sum of Hours:",hour)
        print("sum of minute:",minute)
        print("sum of second:",second)

print("Time of first object")
time1=Time()
print("Time of second object")
time2=Time()
print("-----")
time1.__add__(time2)
print("-----")
```

Output:

Time of first object
Enter the hour:2
Enter the minute:15
Enter the second:07
Time of second object
Enter the hour:4
Enter the minute:37
Enter the second:25

sum of Hours: 6
sum of minute: 52
sum of second: 32
